A New Fast-Switching NBTI Characterization Method That Determines Subthreshold Slope Degradation

Douglas Brisbin and Prasad Chaparala

***Abstract*—For PMOSFET devices, negative bias temperature instability (NBTI) is a serious reliability concern. Because of recovery effects, careful stress and measurement methods must be used to determine threshold voltage degradation. These methods typically assume that mobility and subthreshold slope (SubSlp) degradation are minimal. Recent papers have pointed out that this assumption may not be valid. This paper discusses for the first time a unique fast-switching NBTI measurement technique that alternates between two $V_{\rm GS}$ measurement conditions to determine the SubSlp versus stress time. From these measurements, the effect of SubSlp degradation on V_T degradation can be accurately determined, and results are compared to the standard techniques.**

***Index Terms*—NBTI test methods, negative bias temperature instability (NBTI), PMOSFET reliability.**

I. Introduction

NEGATIVE bias temperature instability (NBTI) is a serious reliability concern in state-of-the-art PMOSFET devices [1]–[11]. The implementation of nitrided gate oxide to prevent boron penetration aggravated the NBTI issue [1], [2], [5]. Because of recovery effects [14]–[17], careful stress and measurement techniques (e.g., "on-the-fly" and "fast $V_{\rm GS}$ switching") must be used to determine threshold voltage degradation [3], [12]–[14]. These methods assume that mobility and subthreshold slope (SubSlp) degradation are minimal. Recent papers have pointed out that this assumption may be invalid [18]–[20]. This paper discusses for the first time a unique NBTI measurement technique that alternates between two $V_{\rm GS}$ test conditions to determine the SubSlp versus stress time. From these measurements, the effect of SubSlp degradation on V_T degradation can be accurately determined, and results are compared to the standard techniques. This paper also introduces an improved NBTI test methodology to complement the pending NBTI JEDEC 14.2.2 testing standard [21].

II. Experimental Details

The devices used in this paper are a 1.2 V PMOS with a $W/L = 10/0.12$ μm and an oxide thickness of 2 nm and a 3 V DGO PMOS with a $W/L = 10/0.3$ μm and an oxide thickness of 6 nm. These devices are tested on a standard Keithley S900 parametric test system using the $V_{\rm GS}$ switching NBTI test method shown in Fig. 1 [21]. In this technique, the initial linear subthreshold I–V behavior of the device is obtained typically at $V_{\rm DS} = -0.05$ V, and the constant current threshold voltage $(V_{T(\rm ci)})$ and initial SubSlp (0) are determined at a target drain current given by

$$I_D = I_{D0} \times W/L, \qquad \text{with } I_{D0} = 50{-}100 \text{ nA}. \tag{1}$$

During the stress interval, the $V_{\rm GS}$ stress is applied (typically with $V_{\rm DS} = 0$ or -0.05 V) for logarithmic half-decade stress times (e.g., 1, 3, 10, 30, 100, etc.). After each stress time, the magnitude of $V_{\rm GS}$ stress is reduced to a fixed $V_{\rm GS}$ at a test value that is equal to the initial $V_{T(\rm ci)}$, and the drain current $I_D(t_{\rm str})$ is measured after a specified time delay (Δ). V_T degradation at stress time $t_{\rm str}$ is determined using

$$\Delta V_T(t_{\rm str}) = \log 10\left(I_D(0)/I_D(t_{\rm str})\right)/\text{SubSlp}(0) \tag{2}$$

where $I_D(0)$ is the initial drain current at $V_{T(\rm ci)}$. Equation (2) is derived assuming that the SubSlp (0) is constant during stress, as shown in Fig. 2. Stress termination occurs when a specified accumulative stress time is reached. At this time, the final I–V behavior and post-stress SubSlp are determined. Fig. 3(a)

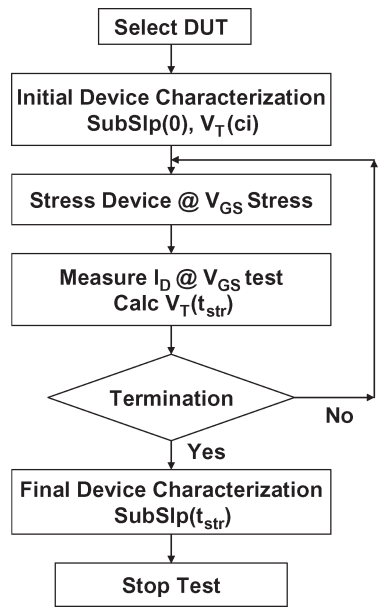

Fig. 1. "Fast-$V_{\rm GS}$-switching" test methodology.

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The authors are with National Semiconductor, Santa Clara, CA 95052 USA (e-mail: douglas.brisbin@nsc.com; prasad.chaparala@nsc.com).

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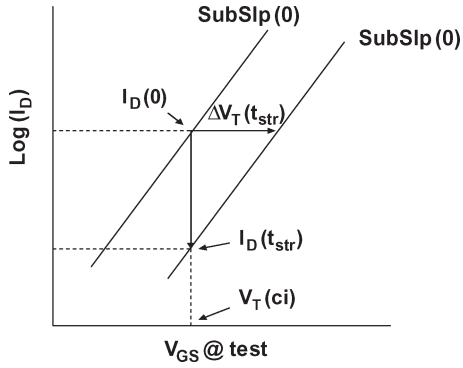

Fig. 2. Method to extract ΔV_T from the measured $I_D(t_{\rm str})$ and the initial SubSlp (0).

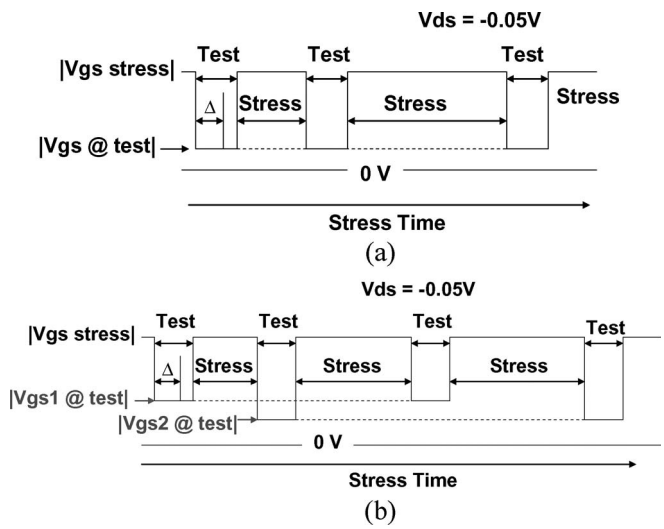

Fig. 3. (a) Fast-$V_{\rm GS}$-switching schematic. (b) New method using alternate $V_{\rm GS}$ test voltages ($V_{\rm GS}1$ and $V_{\rm GS}2$).

shows the stress–measurement time sequence of the proposed JEDEC standard technique [21]. Recent authors [18]–[20] have pointed out that assuming a constant SubSlp (0) may not be valid, and therefore, the $\Delta V_T(t_{\rm str})$ determined by the standard technique may be in error. Unfortunately, making multiple I_D measurements to determine the actual SubSlp after any stress time interval is not accurate because of NBTI recovery effects.

Historically, the SubSlp has typically been reported in terms of a subthreshold swing (SS) as the number of millivolts changes in the $V_{\rm GS}$ required to change the drain current by one decade. SS is simply equal to the reciprocal of the SubSlp ($\rm SS = SubSlp^{-1}$) and has units of millivolts per decade.

This work differs from the standard technique in that two different $V_{\rm GS}$ test conditions are implemented. The first $V_{\rm GS}$ at test is $V_{\rm GS}1 = V_{T(\rm ci)}$, while the second one is $V_{\rm GS}2 = V_{T(\rm ci)} + \Delta V_{\rm GS}$ (typically $\Delta V_{\rm GS} = 20{-}50$ mV). These two $V_{\rm GS}$ at test conditions are applied at alternating stress intervals, as seen in Fig. 3(b). For most of the measurement results shown here, the source SMU was set for single line cycle average, and a 3 ms delay (Δ) to allow for SMU settling time resulted in $\sim$25 ms $V_{\rm GS}$ at test pulse width. The fastest achievable test pulse for the instrumentation used here (no line cycle averaging) was $\sim$4 ms, which included a 1-ms settling delay (Δ).

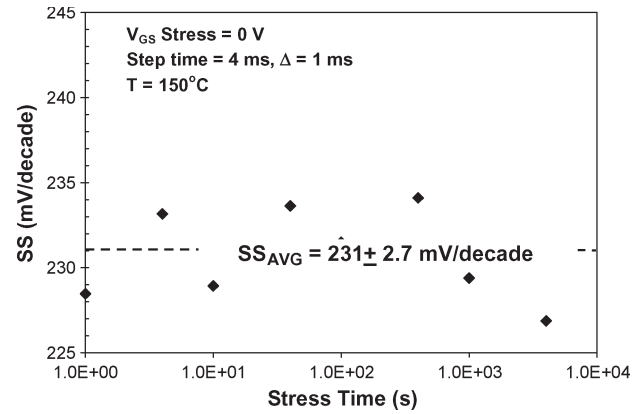

Fig. 4. Measured SS versus stress time for a $V_{\rm GS}$ stress condition of 0 V for the PMOS (3 V) device, showing that the measurement repeatability is $+1.2\%$ (1 sigma).

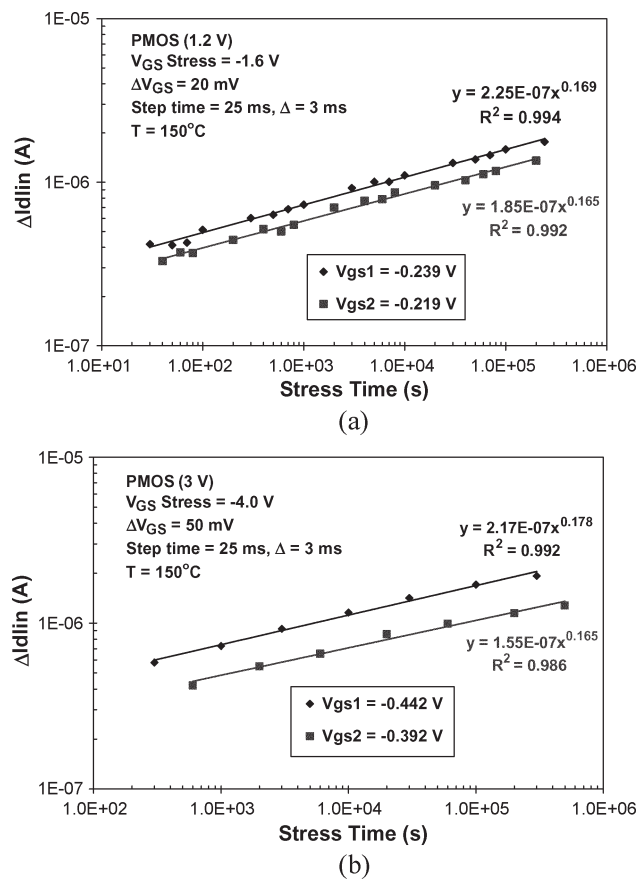

Fig. 5. I_D degradation (ΔI_D) obtained using the two-$V_{\rm GS}$ test method for the (a) PMOS (1.2 V) and (b) PMOS (3.0 V) devices.

A fixed SMU measurement range is implemented during the stress–measurement test sequence to prevent a momentary zero $V_{\rm GS}$ bias during SMU autoranging. After each stress time interval, the $I_D(t_{\rm str})$ is recorded. To validate the test methodology and measurement repeatability, the proposed technique was used to obtain SS in millivolts per decade versus stress time for the $V_{\rm GS}$ stress condition of 0 V (no stress). As seen in Fig. 4, the measurement repeatability is $\sim$1.2%.

Fig. 5 shows typical I_D versus stress time results for the two $V_{\rm GS}$ test method from the PMOS (1.2 V) and DGO PMOS

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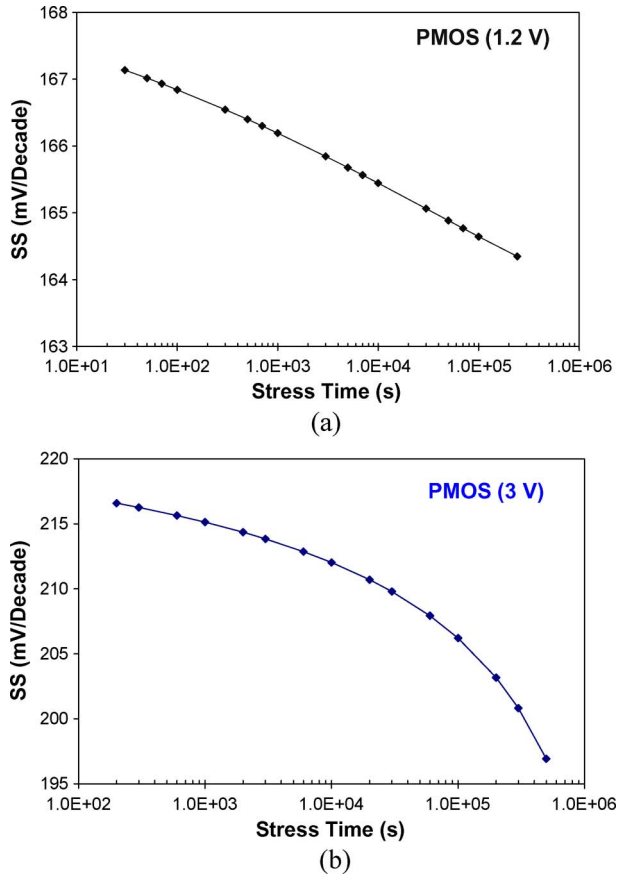

Fig. 6. SS as a function of stress time for the (a) PMOS (1.2 V) and (b) PMOS (3.0 V) devices determined from Fig. 5 data.

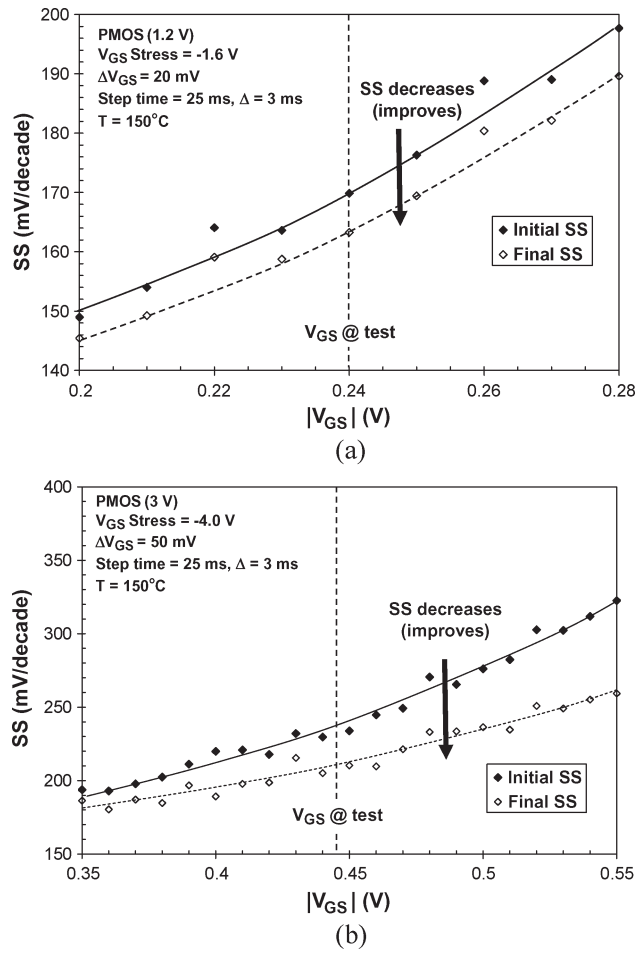

Fig. 7. Initial and post-stress SS versus $V_{\rm GS}$ for the (a) PMOS (1.2 V) and (b) PMOS (3.0 V) devices, showing that SS does indeed improve (decrease) with stress time.

(3 V) devices. In the case of the two $V_{\rm GS}$ test method, the number of measurement iterations was increased to improve the statistical determination of the power-law slope for each $V_{\rm GS}$. As expected, each set of data has power-law time dependence over several time decades. From Fig. 5 data, fits of intermediate values of $\Delta I_D(t_{\rm str})$ can be determined by interpolation at each measurement time. $\mathrm{SS}(t_{\rm str})$ is determined using the following:

$$\mathrm{SS}(t_{\rm str}) = \Delta V_{\rm GS}/\log_{10}\left(I_D1(t_{\rm str})/I_D2(t_{\rm str})\right) \qquad (3)$$

where $I_D1(t_{\rm str})$ and $I_D2(t_{\rm str})$ are the measured and interpolated I_D values at stress time $t_{\rm str}$, respectively. The incremental V_T degradation is calculated using

$$\Delta V_T(t_{\rm str}) = \mathrm{SS}(t_{\rm str}) * \log_{10}\left(I_D(t_{\rm strprev})/I_D(t_{\rm str})\right) \qquad (4)$$

where $I_D(t_{\rm strprev})$ is the measured I_D at the previous stress time interval.

Fig. 6 shows calculated SS [(3) and (4)] versus stress time results from the $I_D(t_{\rm str})$ data for the two devices shown in Fig. 5. The data show that SS for both devices is not constant but decreases (improves) with stress time. A decrease in SS with stress time is unexpected and seems to indicate that the turn-off characteristic of the devices improves with stress. To understand this unexpected result, the initial and post-stress SS data are shown in Fig. 7 (note: the post-stress SS data are affected by NBTI recovery effects). Fig. 7 results show that, at the $V_{\rm GS}$ test condition, SS decreases with stress time, and the decrease in SS is consistent with the results shown in Fig. 6.

It is also apparent from Fig. 7 that SS is not constant with respect to $V_{\rm GS}$, as assumed by (2), but decreases with decreased $V_{\rm GS}$. Fig. 8 shows the initial and post-stress I–V and SS behavior of the PMOS (1.2 V) and PMOS (3 V) devices stressed in Fig. 5. These results show that, because of the NBTI-induced increase in $|V_T|$, the stressed device I_D–$V_{\rm GS}$ curves shift to the right, effectively moving the fixed $V_{\rm GS}$ test point to a steeper part of the SS curve. It is also apparent from Fig. 8 that SS is not a constant until well below the V_T at the target I_D of the device.

Fig. 9 compares PMOS (3 V) V_T degradation results from the single $V_{\rm GS}$ at test using the constant SS (0) approximation and the two $V_{\rm GS}$'s at test method where SS versus stress time is determined at each measurement point. As seen in Fig. 9, the constant SS (0) method systematically overestimates the V_T degradation compared to the two $V_{\rm GS}$'s at test method, yielding significant errors of 10% (6 mV) in 1×10^5 s and 12% (26 mV) in one year. Figs. 6–9 show that for most devices, the constant $V_{\rm GS}$ test method has a measurement flaw since SS typically decreases with stress time resulting in significant and systematic error. One way to resolve this issue is to increment the $V_{\rm GS}$ at test after each stress interval by the measured V_T

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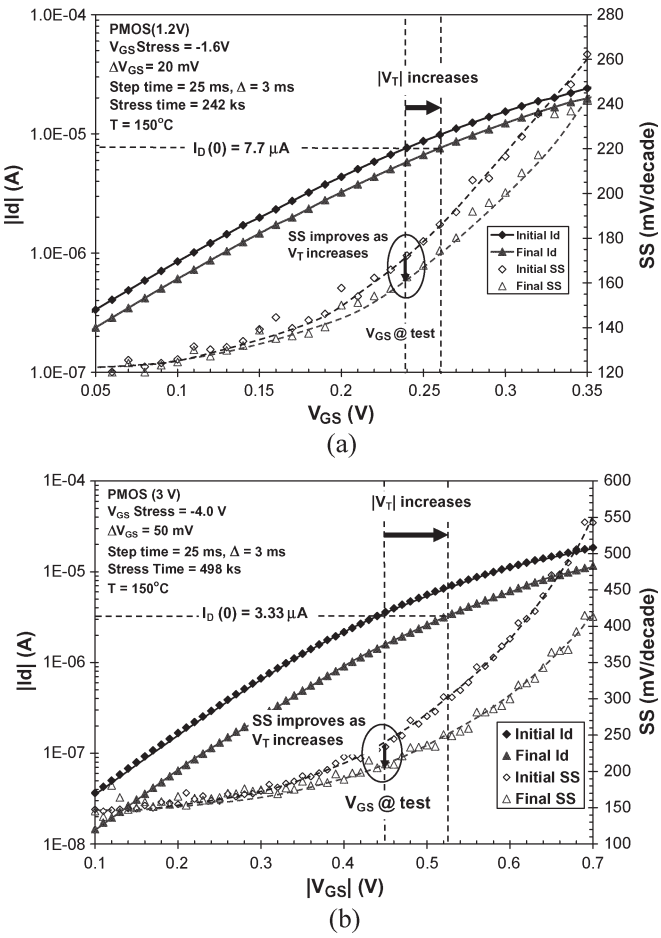

Fig. 8. Cause of the SS improvement in both the (a) PMOS (1.2 V) and (b) PMOS (3 V) devices seen in Fig. 6 is a shift of the I_D–$V_{\rm GS}$ curve to the right during NBTI stress, effectively moving the fixed $V_{\rm GS}$ at test point to a lower $V_{\rm GS}$ value and the steeper part of the stressed I_D–$V_{\rm GS}$ curve.

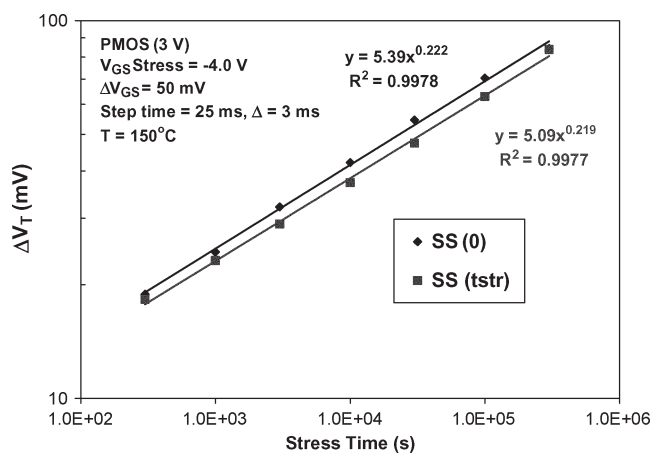

Fig. 9. PMOS (3 V) ΔV_T results from the single $V_{\rm GS}$ at test using the constant SS (0) approximation and the two $V_{\rm GS}$'s at test method where SS$(t_{\rm str})$ is determined at each measurement point.

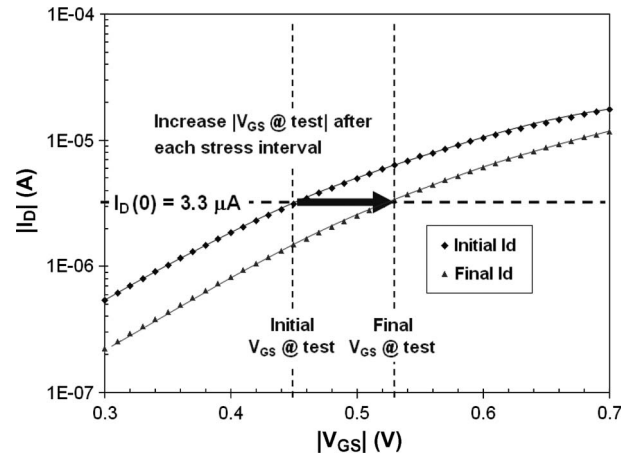

Fig. 10. Incrementing the $V_{\rm GS}$ at test after each stress interval using (2) essentially keeps the $V_{\rm GS}$ at test measurement point at the device $I_D = I_D(0)$ test point.

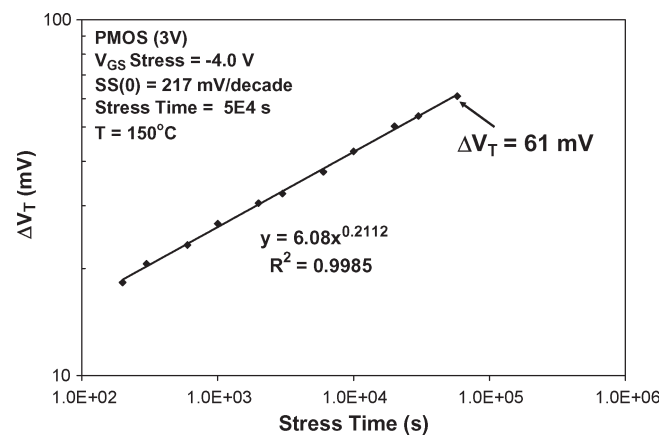

Fig. 11. PMOS (3 V) V_T degradation for a device where the $V_{\rm GS}$ at test is incremented after each stress interval using (2), assuming that SS is constant [SS (0)] during stress.

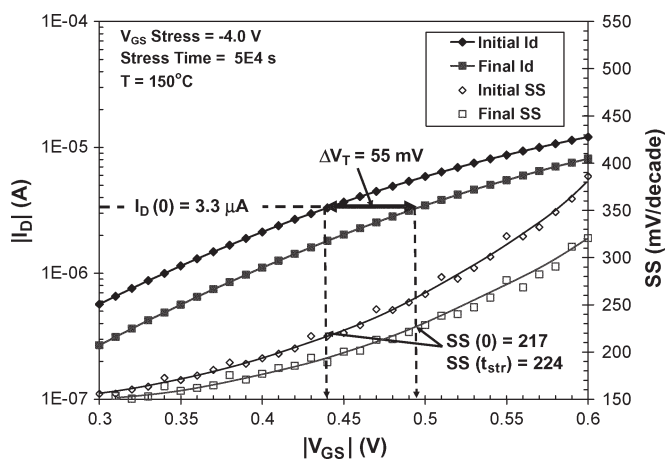

Fig. 12. Pre- and post-stress I_D and SS results for the PMOS (3 V) device stressed in Fig. 11. Results show that the ΔV_T is in qualitative agreement with the value obtained Fig. 11, with SS increasing (degrading) by 3% during stress.

degradation. As seen in Fig. 10, this keeps the $V_{\rm GS}$ at test measurement point at the device $I_D = I_D(0)$ test point. Since the device I_D is returned to $I_D(0)$ after each stress interval, the change in SS due to a decrease in I_D should be minimal.

Fig. 11 shows NBTI degradation results from the PMOS (3 V) device using this method for the case where SS is assumed constant $=$ SS(0) during stress. Fig. 12 shows initial and post-stress I_D and SS versus $V_{\rm GS}$ results from the device

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stressed in Fig. 11. These results show that the total ΔV_T is in qualitative agreement with the value obtained in Fig. 11, with SS increasing (degrading as expected) by $\sim$3% during stress.

III. Summary and Conclusion

A new fast-switching NBTI measurement method was introduced based on applying alternating $V_{\rm GS}$ test conditions to determine SS. The NBTI results using this new method from both PMOS (1.2 V) and PMOS (3 V) devices showed that, unexpectedly, SS decreases (improves) with stress time. It was determined that SS improves with NBTI stress because the device $|V_T|$ increases, effectively moving the fixed $V_{\rm GS}$ at test point to a reduced $|I_D|$ point.

An improvement in the constant-SS test methodology [21] was proposed, which consists of incrementing the test $V_{\rm GS}$ after each stress interval, effectively keeping the test point at $I_D(0)$. The NBTI results from this improved method were in qualitative agreement with the post-stress I_D–$V_{\rm GS}$ results.

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Douglas Brisbin received the B.S. degree from Miami University, Oxford, OH, and the M.S. and Ph.D. degrees in physics from Kent State University, Kent, OH.

Since 2001, he has been with the National Semiconductor, Santa Clara, CA, where he is currently working as the Principal Device Reliability Engineer in the Advanced Process Technology Development Group. His interests include the areas of MOS/bipolar hot-carrier degradation, ultrathin gate oxide integrity, and NBTI.

Dr. Brisbin is an active member of the JEDEC 14.2 Wafer Level Reliability Standards Committee and contributed to more than 30 technical publications.

Prasad Chaparala received the Ph.D. degree in reliability engineering from the University of Maryland, College Park, in 1996.

Since 1996, he has been with the National Semiconductor Corporation, Santa Clara, CA, where he is currently working as the Director of Technology Reliability, responsible for the reliability qualification of new technologies and products. He has contributed to more than 40 publications in international journals/conference proceedings. He is the holder of seven U.S. patents. His areas of interest include analog and mixed signal product reliability degradation mechanisms.

Dr. Chaparala was a recipient of three Best Paper Awards at the IEEE International Reliability Physics Symposium (IRPS). He has served on the management and technical program committees of the IEEE IRPS, International Integrated Reliability Workshop, and International Symposium on the Physical and Failure Analysis of Integrated Circuits. He is currently serving as the General Chair of the IEEE Santa Clara Valley Electron Devices Society Chapter.

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